EXPERIMENT 9

Student Name: Sumit Chauhan **UID: 24BAI70884**
Branch: AIT-CSE **Section/Group: 24AIT_KRG-G2**
Semester: 4
Subject Name: Database Management System **Subject Code: 24CSH-298**

Experiment 9 – Design of a PL/SQL Package with Procedures and Shared Cursor (Enterprise Applications)

Experiment

Designing and implementing a PL/SQL package that contains procedures and a shared cursor to fetch and display employee details. This experiment demonstrates modular programming, code reusability, and efficient data handling using packages in PL/SQL.

Aim

To create and implement a PL/SQL package consisting of a package specification and package body that includes procedures and a shared cursor for retrieving and displaying employee data.

Objective

- To understand the concept and structure of PL/SQL packages
- To differentiate between package specification and package body
- To implement procedures inside packages
- To use shared cursors for efficient data retrieval
- To develop modular and reusable database programs

Software Requirements

Database Management System:
Oracle Database Express Edition (Oracle XE) / PostgreSQL

Database Administration Tool:
Oracle SQL Developer / pgAdmin

Problem Statement
In enterprise-level database applications, related operations must be grouped together for better maintainability, performance, and reusability. A PL/SQL package provides a structured

approach to organize procedures and shared cursors, ensuring efficient data retrieval and modular program design.

Practical / Experiment Steps

1. Create an employee table with relevant attributes
2. Insert sample data into the table
3. Create package specification declaring procedures and cursor
4. Create package body defining logic for procedures and cursor
5. Use a shared cursor to fetch employee records
6. Execute the package procedure
7. Display the output

Procedure

1. Open Oracle SQL Developer and connect to the database
2. Create the employee table
3. Insert sample records
4. Create package specification
5. Create package body
6. Compile the package successfully
7. Execute package procedure
8. Display employee details output

Input / Output Details

Input
Table: employee (id, name, gender, salary)

Step-wise Output

CHANDIGARH UNIVERSITY
Discover. Learn. Empower.

NAAC GRADE A+

EXPERIMENT 9

Step 1 – Create Employee Table

```sql
CREATE TABLE employee (
    id NUMBER PRIMARY KEY,
    name VARCHAR2(50),
    gender VARCHAR2(10),
    salary NUMBER
);
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
SQL> CREATE TABLE employee (
        id NUMBER PRIMARY KEY,
        name VARCHAR2(50),
        gender VARCHAR2(10),...
Show more...


Table EMPLOYEE created.

Elapsed: 00:00:00.024
```

EXPERIMENT 9

Step 2 – Insert Sample Data

```sql
-- This step inserts sample employee records
INSERT INTO employee VALUES (1, 'Amit', 'Male', 50000);
INSERT INTO employee VALUES (2, 'Neha', 'Female', 60000);
INSERT INTO employee VALUES (3, 'Raj', 'Male', 55000);
INSERT INTO employee VALUES (4, 'Priya', 'Female', 65000);

COMMIT;
```

Query result | Script output | DBMS output | Explain Plan | SQL history

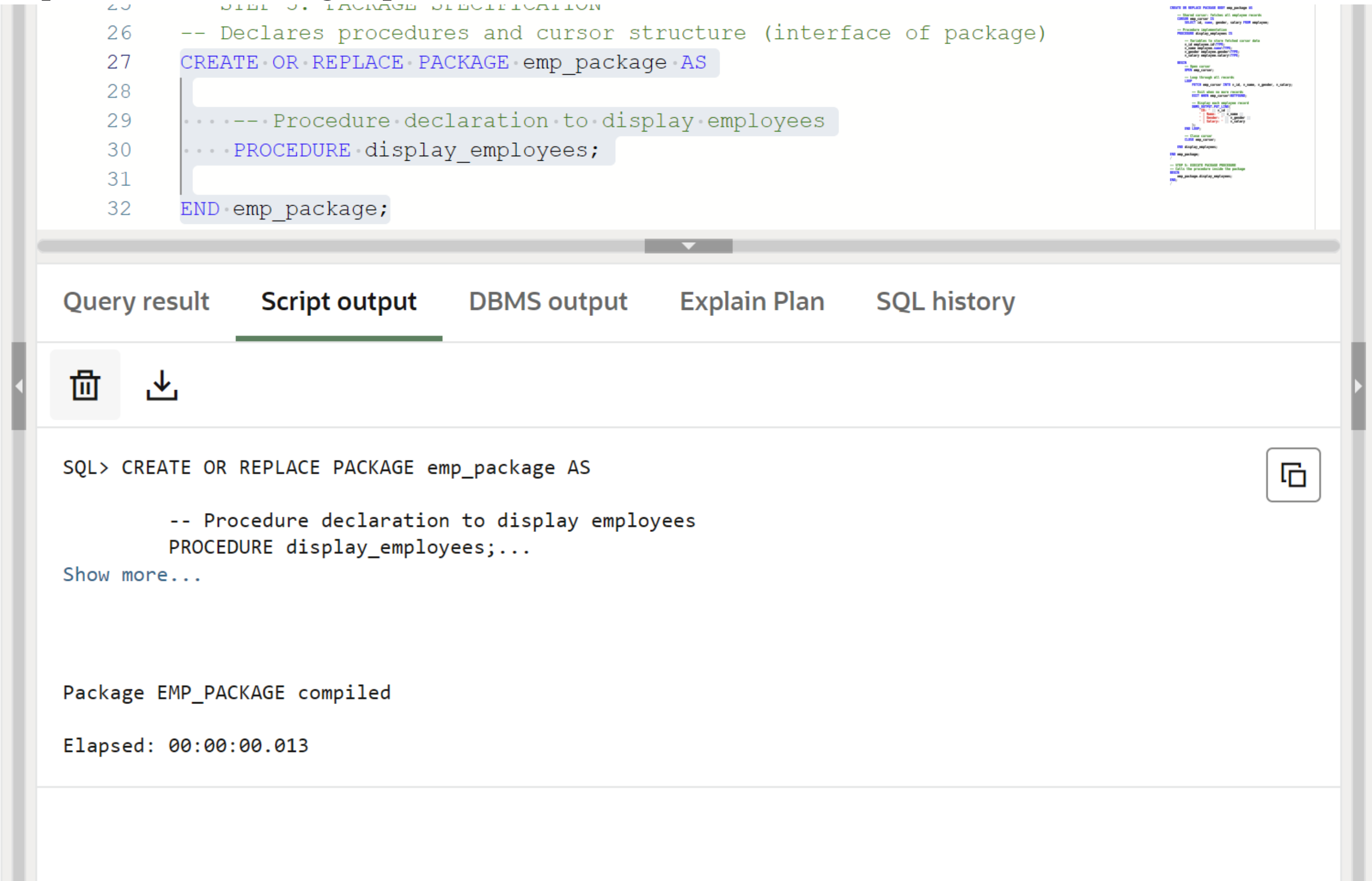
```
SQL> INSERT INTO employee VALUES (4, 'Priya', 'Female', 65000)


1 row inserted.

Elapsed: 00:00:00.002
```

```
SQL> COMMIT


Commit complete.

Elapsed: 00:00:00.002
```

Terms and Conditions | Your Privacy Rights | Delete Your FreeSQL Account | Cookie Preferences

EXPERIMENT 9

Step 3 – Create Package Specification



EXPERIMENT 9

Step 4 – Create Package Body

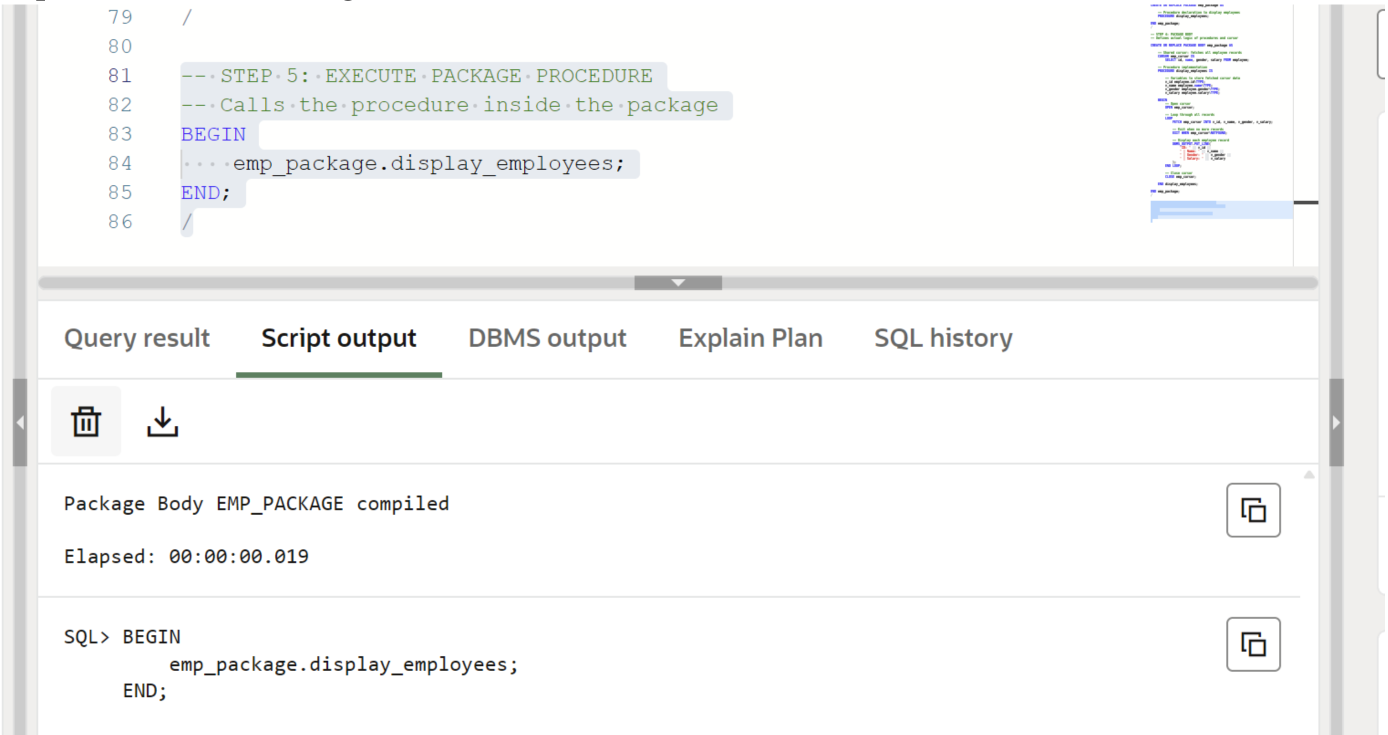
```sql
-- STEP 4: PACKAGE BODY
-- Defines actual logic of procedures and cursor

CREATE OR REPLACE PACKAGE BODY emp_package AS

    -- Shared cursor: fetches all employee records
    CURSOR emp_cursor IS
        SELECT id, name, gender, salary FROM employee;

    -- Procedure implementation
```

Query result | Script output | DBMS output | Explain Plan | SQL history

```
Show more...

Package EMP_PACKAGE compiled

Elapsed: 00:00:00.013

SQL> CREATE OR REPLACE PACKAGE BODY emp_package AS

        -- Shared cursor: fetches all employee records
        CURSOR emp_cursor IS...
Show more...

Package Body EMP_PACKAGE compiled
```

EXPERIMENT 9

Step 5 – Execute Package Procedure



Learning Outcome

After completing this experiment, the learner will be able to:

- Understand the structure of PL/SQL packages
- Create package specification and package body
- Use shared cursors inside packages
- Implement modular and reusable database logic
- Apply package concepts in enterprise applications